



# FRIENDLY AND INCLUSIVE SPACES AWARDS

Category:  
4. Research

Title of Research:

The Autism Friendly University Design Guide

Name of Architect(s):

Magda Mostafa  
Progressive Architects

Abstract of Research:

https://issuu.com/magdamostafa/docs/the\_autism\_friendly\_design\_guide



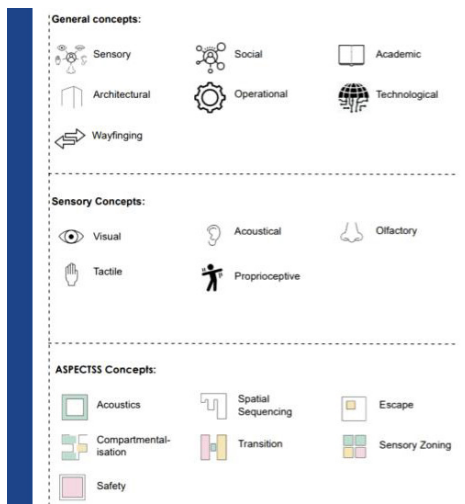
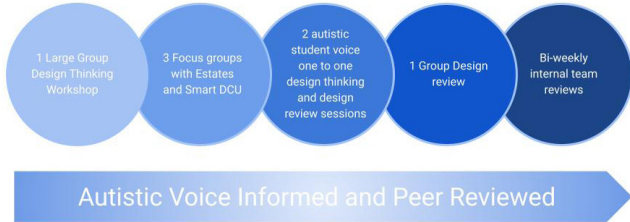
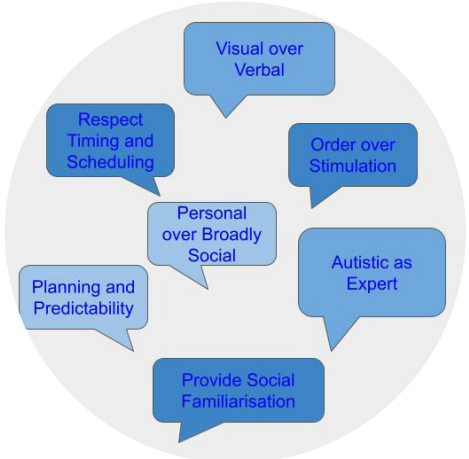
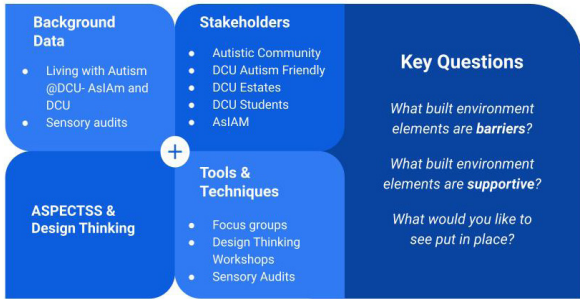
Magda Mostafa, PhD, Autism Design Consultant, Progressive Architects  
Associate Professor of Design, American University in Cairo



METHODOLOGY

Two key sources of data were provided by the University and reviewed to foreground this guide: 1. the 2016 “Living with Autism as a University Student at Dublin City University: Developing an Autism Friendly University” 2. sensory audits carried out by various DCU autism stakeholders- students, advocates and staff. The former resource was framed around the objective of identifying and exploring the current services at DCU targeting autism. It aimed to identify gaps and help improve life on campus for these students. The stakeholders base in the study were broad and included students, staff and faculty across multiple campuses. The methodology used began with a literature review of

peer reviewed research on the topic of higher education support of autism, as well as a web-based review of European and American higher education institutions. This foregrounding research helped in turn inform the 4 studies that comprised the final project- an online survey of DCU students with autism; an online survey of all DCU students; focus groups and semi-structured interviews of academics and support staff; and a sensory audit of two DCU campuses by 4 students with autism with a focus of noise, smells, lighting and signage. Additional sensory audits were carried out by Autism Friendly DCU and shared for the purpose of informing this guide.



OUTCOME & IMPACT

DESIGN THINKING - THE AUTISTIC PERSPECTIVE

We are currently developing translations and versions of multiple Design Thinking tools specific to the autistic community for use during workshops specifically with and for the autistic community.

- Visual over verbal: visual prompts, cues and images can help illustrate ideas and problem spaces in the design thinking workshop. Visual organization of material can be helpful, particularly in orienting and connecting concepts.
- Order over stimulation: typically design thinking sessions may quickly become visually chaotic and over-stimulating with multiple ideas layering and flowing. This can be overwhelming for the autistic participant. Efforts were made to create clear grids of space where specific ideas were to be shared. Separate boards and “fresh starts” were created for each new portion of the workshop.
- Personal over broadly social: participants workshop which create a significant barrier to autistic participation. Refinement of the participant number and organization was carried out in this workshop, with a total of 15 participants split into groups of 5 with one moderator. This allowed a more personal connection among participants and allowed for a manageable social environment.
- Autistic social space balanced with Neurotypical social space and Autistic expertise: the organization of the participants was balanced to allow the comfortable and stigma-free sharing of autistic voice as the “expert” voice. Groupings were curated to include a member from all stakeholder representation, but with a focus on autistic perspective as the expert user.
- Planning ahead and social familiarization: the objective, detailed agenda and participant bios of the workshop were shared ahead of the workshop. A video tutorial illustrating how to use the virtual visual board was also shared, with a contact person for technical support. Time was allocated in the agenda to ensure that everyone was comfortable using the platform and that everyone was familiar with one another, and their role in the workshop.
- Timing and scheduling is key: the shared agenda was organized down to the granularity of 5 minutes, with clear start and stop times for each activity, and clear times for breaks. The activities themselves were timed with a visual timer that everyone could see, a practice common in the Design Thinking method, and very supportive of the autistic perspective.
- Predictability and preparedness: as a general concept, in the planning and process of the workshop, the objectives were always clearly stated, the agenda always visible and recapped often and expectations were always clear. Although some design thinking methods may remain open-ended to encourage lack of conformity of ideas, this lack of structure and expectation may be difficult for autistic users, so expectations are shared with the invitation to propose new ideas.

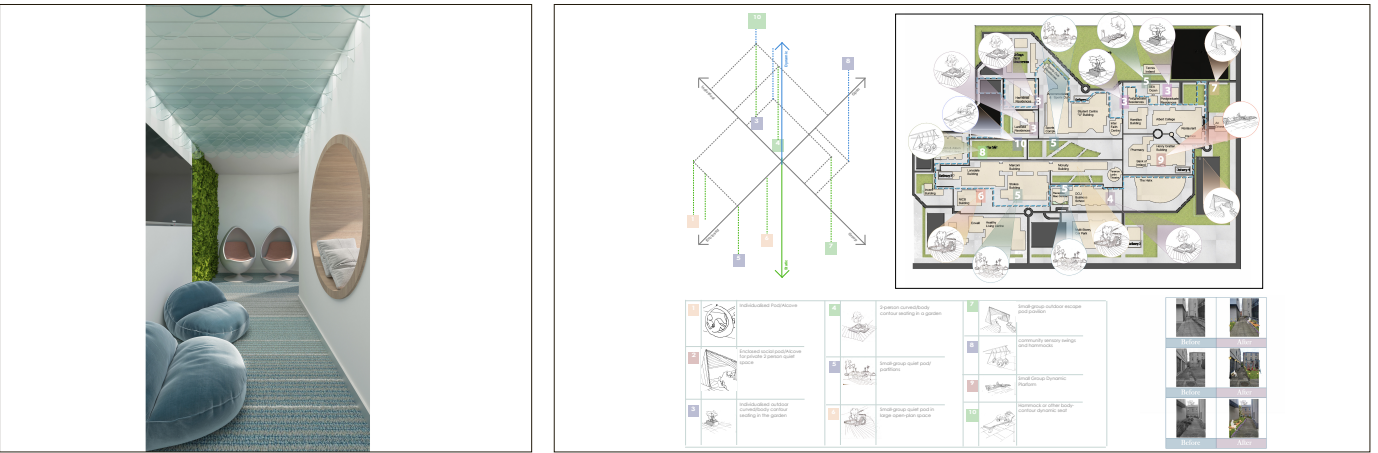
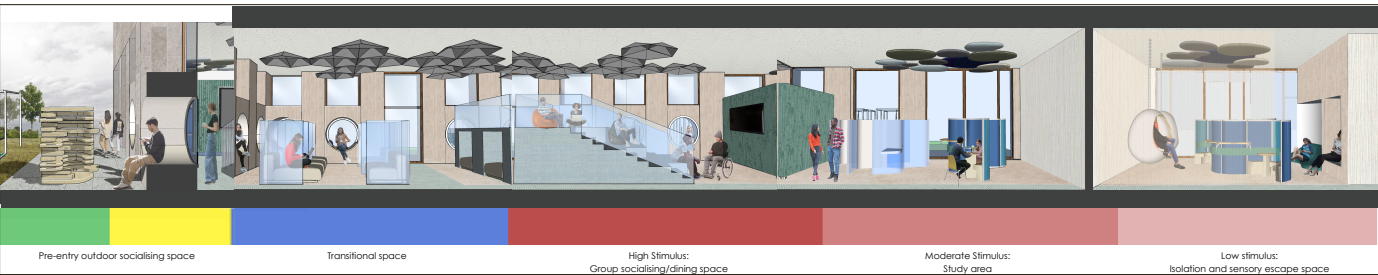
The workshop participants were broadly representative of the DCU Autism Friendly University community including representatives from the autistic student body; the Neurodiverse Student Society; AsIAM Student Ambassadors- Ireland’s National Autism charity and self-advocacy group and DCU partner; DCU Student Services; Occupational Therapists from DCU Student Services; DCU Estates Office and Facilities; DCU Health and Safety; and Smart DCU.

The workshop was moderated by the core design team in partnership with the Coordinator of DCU’s Autism Friendly University Initiative.

PRODUCT

THE AUTISM FRIENDLY DESIGN GUIDE

- Community Integration, Blurring the Boundaries and Space as a Source of Stigma Reduction
- Reverse Inclusion
- Site Planning and Building Design as a Tool for Social and Environmental Management and Mediation
- Integration of Natural Elements and Design of Open Spaces
- Universal Design: the Autistic Perspective
- Agility, Flexibility and Adaptability
- Acoustics
- Spatial Sequencing
- Escape Spaces
- Compartmentalization
- Transition Zones
- Sensory Zoning
- Safety
- Colour
- Lighting
- Material Selection
- Furnishing
- Wayfinding and Navigation
- Technology
- Sensory Economics
- Programming and Operation



INTRODUCTION

HISTORY OF THE INITIATIVE

Dublin City University (DCU) was designated the world’s first autism-friendly university in 2018. It is committed to supporting and celebrating all students and staff who are on the autism spectrum. DCU works closely with global best practices and has committed to adapting the environment, raising awareness and acceptance, and building initiatives to make it as easy as possible for autistic community members to participate fully in all aspects of university life. Pursuant of that goal, they have commissioned the development of this Autism Friendly University Design Guide to provide the requisite built environment infrastructure to realise their programs, processes and procedures to successfully realise a truly autism friendly university.

Autism spectrum disorder diagnoses have been increasing globally in the past two decades, with many scholars and researchers pointing to the early 2000’s as the point where these diagnoses spiked. This was aligned with expanding diagnostic definitions; increasing awareness across the medical, educational and public realms; and the redefinition of some otherwise misunderstood diagnosis as actually autism spectrum.

“many members of this “peak cohort” are now increasingly emerging from their lives as schoolchildren, and as they age out of school, find themselves ready for a higher education that is unfortunately unprepared for them.”

Now, almost 20 years later, what I call this “peak cohort” has progressed through various early intervention, early childhood, Primary and Secondary level mainstream, special education, and alternative format educational systems. Their progress has been supported by research, multiple interventions, evolving pedagogies, growing societal awareness at a level not afforded to the cohorts of the decades preceding them.

Although far from perfect, and with much room for increased awareness, acceptance, inclusion and support, many members of this “peak cohort” are now

“Research has shown that the architectural environment can play a conducive role in the facilitation of inclusion and support of access for autistic individuals.”

increasingly emerging from their lives as schoolchildren, and as they age out of school, find themselves ready for a higher education that is unfortunately unprepared for them. A mindful organization of the built environment of these higher education

spaces may, however, play an important role in achieving these goals, by removing barriers and creating spatial opportunity for more authentic and integrated inclusion of autistic students with their fellow learners. Research has shown that the architectural environment can play a conducive role in the facilitation of inclusion and support of access for autistic individuals, particularly in learning environments. This guide is hopefully a small first step to that preparation at the level of higher education.

APPLICATION & POST OCCUPANCY

AUDIT

Dissemination and replication It is the intent of this guide, that it be used broadly to inform both existing and future designs of higher education campuses across Ireland, Europe and globally. It is an open-access document intended to catalyse design processes at different scales and across different functions of the higher education campus, but also other typologies. It is intended to be replicated across the higher education landscape but also scaled up and across different typologies.

This process may broadly be guided by the following Autism Friendly Design Audit, which can help assess existing environments, evaluate pilot designs and provide gap analysis to help prioritise facilities and design decisions.

| Area             | Category  | Item                    | Notes                                                                     |
|------------------|-----------|-------------------------|---------------------------------------------------------------------------|
| General Concepts | Sensory   | 1. Visual               | Visual clutter, visual noise, visual distractions                         |
|                  |           | 2. Auditory             | Auditory clutter, auditory noise, auditory distractions                   |
|                  |           | 3. Tactile              | Tactile clutter, tactile noise, tactile distractions                      |
|                  |           | 4. Olfactory            | Olfactory clutter, olfactory noise, olfactory distractions                |
|                  |           | 5. Gustatory            | Gustatory clutter, gustatory noise, gustatory distractions                |
|                  |           | 6. Vestibular           | Vestibular clutter, vestibular noise, vestibular distractions             |
|                  |           | 7. Proprioceptive       | Proprioceptive clutter, proprioceptive noise, proprioceptive distractions |
|                  |           | 8. Interoceptive        | Interoceptive clutter, interoceptive noise, interoceptive distractions    |
|                  |           | 9. Cognitive            | Cognitive clutter, cognitive noise, cognitive distractions                |
|                  |           | 10. Emotional           | Emotional clutter, emotional noise, emotional distractions                |
| Sensory Concepts | Visual    | 1. Visual               | Visual clutter, visual noise, visual distractions                         |
|                  |           | 2. Auditory             | Auditory clutter, auditory noise, auditory distractions                   |
|                  |           | 3. Tactile              | Tactile clutter, tactile noise, tactile distractions                      |
|                  |           | 4. Olfactory            | Olfactory clutter, olfactory noise, olfactory distractions                |
|                  |           | 5. Gustatory            | Gustatory clutter, gustatory noise, gustatory distractions                |
|                  |           | 6. Vestibular           | Vestibular clutter, vestibular noise, vestibular distractions             |
|                  |           | 7. Proprioceptive       | Proprioceptive clutter, proprioceptive noise, proprioceptive distractions |
|                  |           | 8. Interoceptive        | Interoceptive clutter, interoceptive noise, interoceptive distractions    |
|                  |           | 9. Cognitive            | Cognitive clutter, cognitive noise, cognitive distractions                |
|                  |           | 10. Emotional           | Emotional clutter, emotional noise, emotional distractions                |
| ASPECTS Concepts | Acoustics | 1. Acoustics            | Acoustic clutter, acoustic noise, acoustic distractions                   |
|                  |           | 2. Spatial Sequencing   | Spatial clutter, spatial noise, spatial distractions                      |
|                  |           | 3. Escape Spaces        | Escape clutter, escape noise, escape distractions                         |
|                  |           | 4. Compartmentalization | Compartment clutter, compartment noise, compartment distractions          |
|                  |           | 5. Transition Zones     | Transition clutter, transition noise, transition distractions             |
|                  |           | 6. Safety               | Safety clutter, safety noise, safety distractions                         |
|                  |           | 7. Colour               | Colour clutter, colour noise, colour distractions                         |
|                  |           | 8. Lighting             | Lighting clutter, lighting noise, lighting distractions                   |
|                  |           | 9. Material Selection   | Material clutter, material noise, material distractions                   |
|                  |           | 10. Furnishing          | Furnishing clutter, furnishing noise, furnishing distractions             |

SENSORY VISUALIZATION



Images by Stuart Neilsen, Post-Occupancy Researcher, DCU

EPILOGUE

Higher education campuses have throughout our modern history been the sites of contemplation, scientific discovery and paradigm shifting debate that has shaped our society and moved human progress forward. There is no doubt that such progress was not, nor will be possible without different ways of thinking- the kinds of ways of thinking that neurodiversity brings. It is diversity that provides the fertile ground for the critical discourse that drives this progress. It is diversity that provides the unique and innovative lenses through which the world can be observed, problems approached and solutions tackled to make often field-shifting discoveries. Without diversity our knowledge will stagnate, our academic minds will not be challenged to adapt, grow and develop to create the solutions for our unknown future. But such diversity cannot be effective and productive if it is not supported by a holistic ecosystem of support, facilitation and accommodation. DCU has taken strides towards creating this ecosystem of academic, social and operational support, and with this guide it has expanded those supports to include the built environment- it has now set the stage for that diversity to flourish. Our hopes are that this guide will help complete the holistic landscape for diversity to flourish, and for Neurodiverse and autistic students to find a home, to not only be included into, but to contribute to the paradigm shifting knowledge production needed in our future. We hope that this guide will provide a template to be scaled up to neighborhoods and cities and scaled across to other European nations and across the world, for an autism friendly world, where diversity can flourish everywhere.